

Build Amazon VPC with Public and Private Subnets (from Scratch)

Objective

1. Learn how to build Public and Private subnets from scratch.
2. The VPC wizard will not be used. Every component required to build public and private subnets will be created and configured manually.
3. This will give an in-depth understanding of internal components of VPCs and subnets.

Lab Details

1. This lab walks you through AWS Virtual Private Cloud (VPC) creation from scratch.
2. In this lab, we will create a VPC without using VPC Wizard. We will create our own public and private subnets. Along with that, we will be configuring Route tables.
3. Duration: **30 minutes**
4. AWS Region: **US East (N. Virginia) us-east-1**

Introduction

What is VPC?

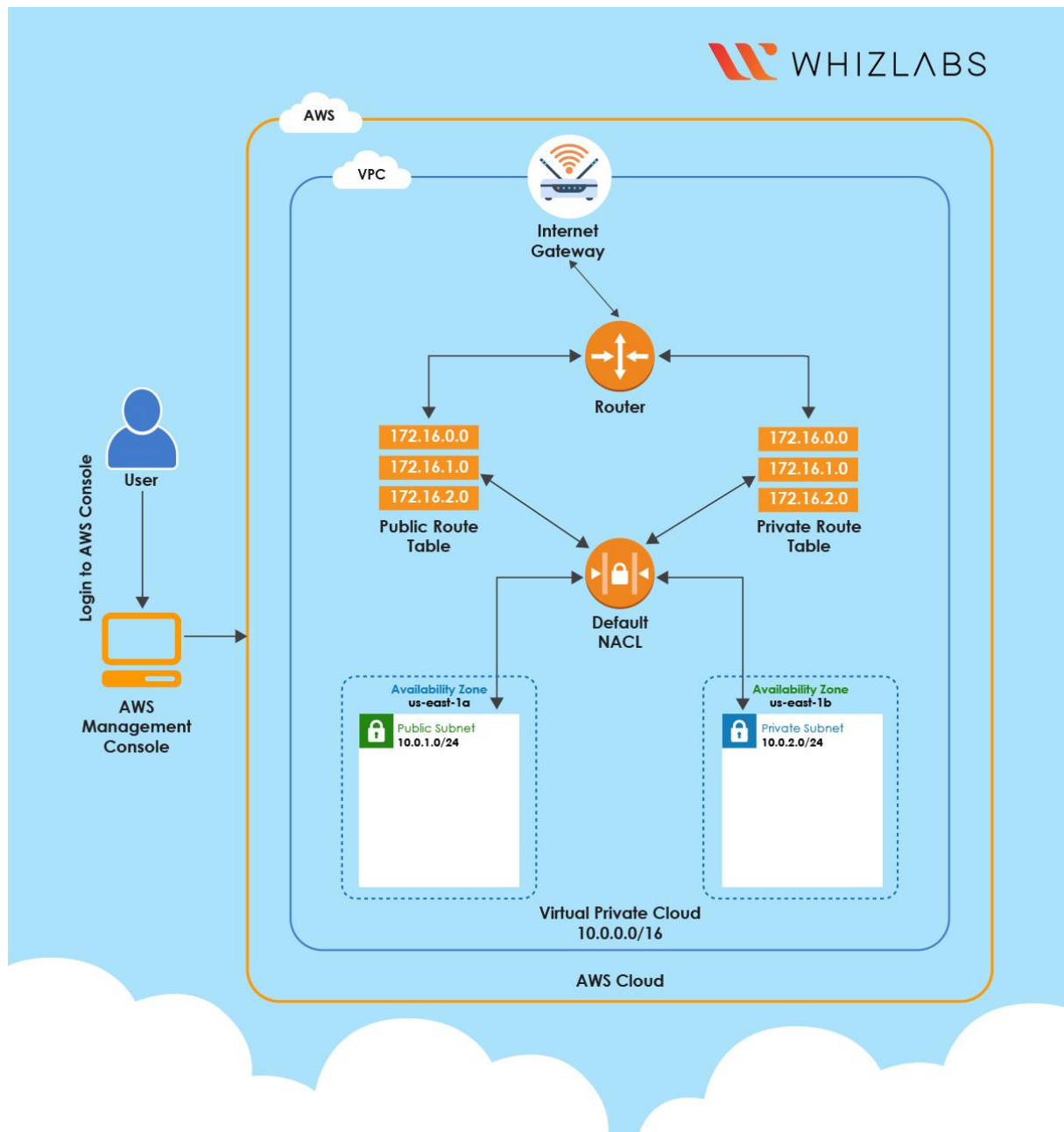
- VPC stands for Virtual Private Cloud.
- It's a custom-defined virtual network within the AWS Cloud.
- Users can logically create their personal network, designing and implementing a separate and independent network that would operate in the AWS Cloud.
- Primary components are : Subnets, IP addresses, NAT Devices (Instances & Gateways), Route Tables, Internet & Virtual Private Gateways, Access Control Lists, Security groups, VPC Endpoints.
- A subnet is a segment of the VPC IP address range, where we can launch EC2 Instances, RDS, and other AWS resources.
- Subnet are further classified as Public and Private.
- Public subnets hold resources that can be accessed from the Internet.
- Common attributes for instances in Public Subnets to have are:
 - Elastic IP (EIP) address or Public IP address attached to the EC2 instance.
 - IGW attached to the VPC.

- The subnet must have a route table entry with destination as internet gateway (IGW) 0.0.0.0/0.
- Security groups and NACLs should not block remote access.
- Public subnets are associated with a route table that directs subnet traffic to the internet using an Internet Gateway.
- Private subnets hold resources that can be accessed from within the VPC network.
- Multiple subnets can be associated with a single route table. However, a single subnet cannot be associated with multiple route tables.
- Route tables hold sets of rules, called **routes** that are used to determine where the traffic is directed.
- Every subnet in a VPC is linked to the route table.
- Primary or Main route tables are the ones that automatically come with your VPC. They control the routing for all subnets that are not explicitly associated with any other route table.
- The default route table cannot be deleted.
- Custom route tables are the ones you create for your VPC, and you can add routes as needed.
- Custom route tables can be deleted when not required.
- Internet Gateway (IGW) is a virtual router which helps a VPC connect to the Internet.
- By default, instances that are launched in a VPC cannot communicate with the Internet. To enable Internet access, Internet gateway needed to be attached to the VPC.
- Public subnets gets connected to IGW through route tables to get accessed over the Internet.
- Internet Gateways are horizontally scalable, highly available and redundant.
- EIP, Elastic IP address is a static IPv4 address used by AWS to manage its dynamic cloud computing services.
- It is associated with an AWS Account, and you can use it to mask if an instance failure occurs i.e., if a server fails, we can map this IP address to another server and keep moving without any issues.
- NAT devices can be either an Instance / Gateway residing in Public subnet, (to which an EIP is assigned).
- NAT devices help instances in Private subnets interact with the Internet.
- Access Control List (ACL) is an optional layer of security that acts as a firewall for controlling network traffic in and out of the subnet.
- Rules are defined with the ACL for allowing or denying network traffic either on ports / IP addresses.

Basic Understanding before we start building VPC from scratch

- When you create an Amazon AWS VPC, you specify a set of IP addresses in the form of a **Classless Inter-Domain Routing (CIDR)** block (Ex: **10.0.0.0/16**).
- You can assign a single CIDR block to a VPC. The allowed block size is between a /28 netmask and /16 netmask. In other words, the VPC can contain from 16 to 65,536 IP addresses.

Architecture Diagram

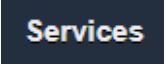


Task Details

1. Log into AWS Management Console.
2. Create a VPC without using the VPC Wizard.
3. Create an Internet Gateway.
4. Create private and public subnets for the VPC.
5. Create and Configure Route tables.
6. Validation of the lab.

Lab Steps

Task 1: Creating New VPC

1. Make sure you are in the **US East (N. Virginia) us-east-1** Region.
2. Navigate to VPC by clicking on  on the top of AWS Console.
3. Click on **VPC** (under **Networking & Content Delivery** section) or you can also search for VPC.
4. Click on **Your VPCs** from the left menu.
5. Here you can see the list of all VPC, No need to do anything with the existing and default VPCs, we will create a new VPC for this lab.
6. Click on .
 - **Name tag:** Enter a VPC name for identification to your VPC. Ex: **MyVPC**
 - **IPv4 CIDR block:** Enter **10.0.0.0/16**
 - **IPv6 CIDR block:** No need to change this, make sure **No IPv6 CIDR Block** is checked.
 - **Tenancy:** No need to change this, make sure **Default** is selected.

- Now click on **Create VPC**.

VPC settings

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

MyVPC

IPv4 CIDR block [Info](#)

10.0.0.0/16

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)

Default

- Once VPC is created, it will appear with details as shown below:

Name	VPC ID	State	IPv4 CIDR
MyVPC	vpc-0aff084cd082b37ec	✓ Available	10.0.0.0/16

Task 2: Creating Subnets

In this lab, we will create one public subnet and a private subnet in us-east-1a and us-east-1b Availability Zones respectively as follows:

- For the Public Subnet, click on **Subnets** from the left menu and click on **Create subnet**.
 - VPC ID** : Select **MyVPC** from the list you created earlier.
 - Subnet Name** : Enter Name **MyPublicSubnet**
 - Availability Zone** : Select **us-east-1a**
 - IPv4 CIDR block** : Enter the range **10.0.1.0/24**
 - Click on **Create subnet**.
- For the Private Subnet, click on **Create subnet** again.
 - VPC ID** : Select **MyVPC** from the list you created earlier.
 - Subnet Name** : Enter Name **MyPrivateSubnet**
 - Availability Zone** : Select **us-east-1b**
 - IPv4 CIDR block** : Enter the range **10.0.2.0/24**

- Click on **Create subnet**.

Subnets (8) [Info](#)

🔍 *Filter subnets*

☐	Name ▼	Subnet ID ▼	State ▼	VPC ▼
☐	–	subnet-5d047f01	✔ Available	vpc-3418934e Default VPC
☐	MyPublicSubnet	subnet-025aec644b5a387e4	✔ Available	vpc-0bc6ac5e4082bc2ba My...
☐	–	subnet-6d090727	✔ Available	vpc-3418934e Default VPC
☐	–	subnet-afb73591	✔ Available	vpc-3418934e Default VPC
☐	MyPrivateSubnet	subnet-06eb435482cdf40f4	✔ Available	vpc-0bc6ac5e4082bc2ba My...
☐	–	subnet-0141710e	✔ Available	vpc-3418934e Default VPC
☐	–	subnet-e54702cb	✔ Available	vpc-3418934e Default VPC
☐	–	subnet-d52759b2	✔ Available	vpc-3418934e Default VPC

Task 3: Create and configure Internet Gateway

1. Click on **Internet Gateways** from the left menu and click

Create internet gateway

- **Name Tag** : Enter ***MyInternetGateway***.

- Click on **Create internet gateway**.

VPC > Internet gateways > Create internet gateway

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	MyInternetGateway	Remove

You can add 49 more tags.

2. Select the Internet gateway you created from the list
 - Click on **Actions**.
 - Click on **Attach to VPC**.

Actions ▲

Attach to VPC

- Select MyVPC which you created from the list and click on **Attach internet gateway**.

Attach to VPC (igw-07c68ed0f807e819a) [Info](#)

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

vpc-0aff084cd082b37ec - MyVPC

► [AWS Command Line Interface command](#)

Cancel

Attach internet gateway

Task 4: Create Route Tables

1. Go to **Route Tables** from the left menu and click on **Create route table**.
 - **Name Tag:** Enter ***PublicRouteTable***.
 - **VPC:** Select **MyVPC** from the list.
 - Click on **Create route table**.
2. Repeat the same steps to create a route table for the Private subnet.
 - **Name Tag:** Enter ***PrivateRouteTable***.
 - **VPC:** Select **MyVPC** from the list.
 - Click on **Create route table**.

Route tables (4) Info		
Filter route tables		
☐	Name ▾	Route table ID ▾
☐	PublicRouteTable	rtb-0e074f14fba65df12
☐	–	rtb-5e859321
☐	–	rtb-021227bf4fdac637a
☐	PrivateRouteTable	rtb-0bbfa28ef81cec6e8

- Now **associate the subnets** to the route tables.
- Click on one **PublicRouteTable** and go to the **Subnet Associations** tab.
 - Click on **Edit subnet associations**.
 - Select **MyPublicSubnet** from the list.
 - Click on **Save associations**.
- Click on one **PrivateRouteTable** and go to the **Subnet Associations** tab.
 - Click on **Edit subnet associations**.
 - Select **MyPrivateSubnet** from the list.
 - Click on **Save associations**.
- Make sure not to associate any subnets with the **Main Route Table**.
- PublicRouteTable**: Add a route to allow Internet traffic to the VPC.
 - Select **PublicRouteTable**.
 - Go to **Routes** tab, click on **Edit routes** and on the next page, click on **Add route**.
 - Specify the following values:
 - Destination**: Enter **0.0.0.0/0**
 - Target**: Select **Internet Gateway** from the dropdown menu to select **MyInternetGateway**.
 - Click on **Save changes**.

Completion and Conclusion

- You have successfully completed and learned how to create public and private subnets.
- You have learned how to create an Internet Gateway and associate it to VPC.
- You now understand how public and private subnets are different (through associating Internet Gateway or not).
- The instances launched inside the public subnet will be able to access the internet and instances launched inside the private subnet will not have access to the internet.